

PINN 1D (hard constraints)

Problem 1: Standard Sine Wave

$$\text{PDE: } -u'' = \pi^2 * \sin(\pi * x)$$

$$\text{Exact: } u(x) = \sin(\pi * x)$$

Problem 2: Simple Polynomial

$$\text{PDE: } -u'' = 2$$

$$\text{Exact: } u(x) = x * (1 - x) \text{ [or } x - x^2]$$

Problem 3: Higher Frequency Sine (Harder for PINNs)

$$\text{PDE: } -u'' = 4 * \pi^2 * \sin(2 * \pi * x)$$

$$\text{Exact: } u(x) = \sin(2 * \pi * x)$$

Problem 4: Non-Zero Boundary Conditions

$$\text{PDE: } -u'' = -e^x$$

$$\text{Exact: } u(x) = e^x$$

$$\text{Boundary: } u(0) = 1, u(1) = e$$

